

VMC-3000 Series | MID 134 (CLS) | MID 136 (LUB) | MID 138 (HYD)

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1. Coolant System (CLS, MID 134)

1.1 System Overview

The VMC-3000 coolant system delivers semi-synthetic metalworking fluid to the cutting zone via flood nozzles and through-spindle coolant (TSC). The system comprises a 300 L stainless steel tank, a 2.2 kW centrifugal pump, air-cooled chiller (setpoint 20 °C), inline flow meter (P030), pressure transducer (P032), temperature sensor (P031), and a level float switch (P033).

WARNING: Machine will perform an emergency stop (CLS-CR-001) if coolant flow P030 drops below 5 L/min during active cutting. Do not bypass the flow switch during production.

1.2 Coolant Specification

Parameter	Specification
Fluid Type	Semi-synthetic metalworking emulsion
Recommended Product	Blaser Swissslube Blasocut 4000 Strong or equivalent
Concentration (aluminium)	6–8% by volume (refractometer reading 3–4% Brix)
Concentration (steel)	8–10% by volume (4–5% Brix)
pH Range	8.5 – 9.5 (check weekly with pH strip)
Maximum Temperature	35 °C (alarm at 40 °C — CLS-MJ-001)
Tank Volume	300 L (fill to 85% operating level)
Replacement Interval	Every 6 months or when concentration/pH out of range
Disposal	Licensed waste coolant disposal only (consult local regulations)

Table 1-1: Coolant Specification

1.3 Coolant System Parameters

Param ID	PID	Parameter Name	Unit	Normal Range	Critical Range
P030	30	Coolant Flow Rate	L/min	15 – 40	5 – 50
P031	31	Coolant Temperature	°C	15 – 35	5 – 45
P032	32	Coolant Pressure	bar	4 – 8	1 – 12
P033	33	Coolant Tank Level	%	20 – 100	5 – 100

Table 1-2: CLS Monitored Parameters

1.4 Coolant Error Codes

Error Code	Severity	Title	Primary Param	Action
CLS-CR-001	CRITICAL	Coolant Flow Loss — Emergency Stop	—	Immediate machine stop. Do not restart with
CLS-MJ-001	MAJOR	Coolant Overtemperature	—	Stop current operation. Machine must be in
CLS-SR-001	SERIOUS	Coolant Pressure — Low	—	Complete current cycle then stop. Schedul

Error Code	Severity	Title	Primary Param	Action
CLS-MD-001	MODERATE	Coolant Tank Level — Low	—	Log fault. Schedule maintenance within 24
CLS-MN-001	MINOR	Coolant Flow — Minor Deviation	—	Log fault. Monitor parameters. Schedule in
CLS-WN-001	WARNING	Coolant Tank — Refill Reminder	—	Log event. No immediate action required. R
CLS-NC-001	NOTICE	Coolant Temperature Stabilised	—	Informational. Parameter approaching limit
CLS-AD-001	ADVISORY	Coolant — Through-Spindle Active	—	Performance advisory only. No maintenanc

Table 1-3: CLS Error Codes

1.5 Coolant Maintenance Schedule

Interval	Task	Error Code Ref
Daily	Check tank level P033 (min 25%); add pre-mixed coolant if low	CLS-WN-001
Weekly	Measure concentration with refractometer; record Brix reading	CLS-MD-001
Weekly	Measure pH with calibrated pH meter; record value	—
Weekly	Check coolant temperature P031 at start and end of shift	CLS-MJ-001
Monthly	Clean coolant filter element; inspect for chips	CLS-SR-001
Monthly	Check coolant pump current (compare to motor nameplate)	—
Monthly	Inspect coolant nozzle positions and alignment	CLS-MN-001
3-monthly	Clean coolant tank: pump out, scrub, refill	DOC-EIQ-004
6-monthly	Full coolant replacement; dispose of old coolant per regulations	—
6-monthly	Inspect chiller condenser fins; clean with fin comb	CLS-MJ-001

Table 1-4: Coolant System Maintenance Schedule

2. Lubrication System (LUB, MID 136)

2.1 System Overview

The centralised lubrication system is a Bijur Delimon-type oil-pulse (progressive distributor) system. An electric pump (0.18 kW) delivers measured pulses of ISO VG 32 way lube oil to all ball screw nuts, linear guide carriages, and ball screw support bearings on a timed cycle (default 15 minutes interval, 2-second pump duration). The system monitors pump pressure (P040), reservoir level (P041), and pump motor current (P042).

NOTE: The lubrication system is the single most important factor in ball screw and guide rail longevity. Failure to maintain oil level P041 above 25% will trigger LUB-SR-001 and accelerate wear, leading to axis following errors (AXS-MJ-001).

Parameter	Specification
Oil Type	ISO VG 32 Way Lube (mineral-based)
Recommended Product	Mobil Vactra Oil No.1 or equivalent
Reservoir Volume	2.5 L
Pump Cycle Interval	15 min (adjustable via parameter P-LUB-01)
Pump Duration	2 s per cycle (adjustable via P-LUB-02)
Nominal Delivery Pressure	2.0–4.5 bar (P040)
Critical Low Pressure	0.5 bar (triggers LUB-CR-001 — machine stop)
Pump Motor	0.18 kW, 3-phase, 50 Hz
Distributor Type	Progressive (SSV metering valves)
Number of Lube Points	18 (8 guide carriages, 6 ball screw nuts, 4 support bearings)

Table 2-1: Lubrication System Specification

2.2 Lubrication Parameters

Param ID	PID	Parameter Name	Unit	Normal Range	Critical Range
P040	40	Lubrication Oil Pressure	bar	2.0 – 4.5	0.5 – 6.0
P041	41	Lubrication Oil Level	%	25 – 100	10 – 100
P042	42	Lubrication Pump Current	A	0.5 – 2.5	0 – 4.0

Table 2-2: LUB Monitored Parameters

2.3 Lubrication Error Codes

Error Code	Severity	Title	Primary Param	Action
LUB-CR-001	CRITICAL	Lubrication System Failure — Machine Stopped	—	Immediate machine stop. Do not restart with
LUB-MJ-001	MAJOR	Lubrication Pump — No Flow Confirmation	—	Stop current operation. Machine must be in
LUB-SR-001	SERIOUS	Lubrication Oil Level — Low	—	Complete current cycle then stop. Schedule
LUB-MD-001	MODERATE	Lubrication Pressure — Marginal	—	Log fault. Schedule maintenance within 24

Error Code	Severity	Title	Primary Param	Action
LUB-MN-001	MINOR	Lubrication Pump Current — Elevated	—	Log fault. Monitor parameters. Schedule in
LUB-WN-001	WARNING	Lubrication — Scheduled Interval Due	—	Log event. No immediate action required. F
LUB-NC-001	NOTICE	Lubrication Cycle Complete	—	Informational. Parameter approaching limit
LUB-AD-001	ADVISORY	Lubrication Oil Level — Nominal	—	Performance advisory only. No maintenanc

Table 2-3: LUB Error Codes

3. Hydraulic System (HYD, MID 138)

The hydraulic system powers workpiece clamping fixtures and the ATC arm swing cylinder. A gear pump (8 L/min, 70 bar nominal) draws from a 20 L reservoir. System pressure is regulated by a pilot-operated relief valve and monitored by P050. Oil temperature is controlled below 55 °C by an air-cooled heat exchanger.

Param ID	PID	Parameter Name	Unit	Normal Range	Critical Range
—	50	Hydraulic System Pressure	bar	60 – 80	40 – 100
—	51	Hydraulic Oil Temperature	°C	30 – 55	20 – 70
—	52	Hydraulic Pump Flow	L/min	20 – 35	5 – 40

Table 3-1: HYD Monitored Parameters

Error Code	Severity	Title	Primary Param	Action
HYD-MJ-001	MAJOR	Hydraulic Pressure — Low	—	Stop current operation. Machine must be in
HYD-SR-001	SERIOUS	Hydraulic Oil Overtemperature	—	Complete current cycle then stop. Schedul
HYD-MD-001	MODERATE	Hydraulic Pressure — Minor Drift	—	Log fault. Schedule maintenance within 24
HYD-WN-001	WARNING	Hydraulic Pump — Scheduled Service Due	—	Log event. No immediate action required. F

Table 3-2: HYD Error Codes

4. Spare Parts — Fluid Systems

Part Number	Description	Qty	Unit	Lead Time	Safety Stock
EIQ-CLS-PUMP-22	Coolant pump 2.2 kW centrifugal (complete)	1	Each	3 weeks	1
EIQ-CLS-FILTER-10	Coolant filter element 10 micron (pack of 10)	1	Pack	1 week	2
EIQ-CLS-FLOW-001	Coolant flow meter (P030 sensor)	1	Each	2 weeks	1
EIQ-CLS-TEMP-001	Coolant temperature sensor NTC (P031)	1	Each	1 week	2
EIQ-LUB-PUMP-018	Lubrication pump 0.18 kW complete unit	1	Each	3 weeks	1
EIQ-LUB-SSV-6	Progressive distributor SSV-6 metering block	1	Each	2 weeks	2
EIQ-LUB-OIL-VG32	ISO VG32 way lube oil, 5 L container	1	Container	3 days	4
EIQ-HYD-PUMP-8	Hydraulic gear pump 8 L/min 70 bar	1	Each	4 weeks	1
EIQ-HYD-SEAL-KIT	Hydraulic cylinder seal kit (all circuits)	1	Kit	1 week	2

Part Number	Description	Qty	Unit	Lead Time	Safety Stock
EIQ-HYD-OIL-HM46	Hydraulic oil ISO VG 46, 20 L	1	Can	3 days	1

Table 4-1: Coolant/Lubrication/Hydraulic Spare Parts